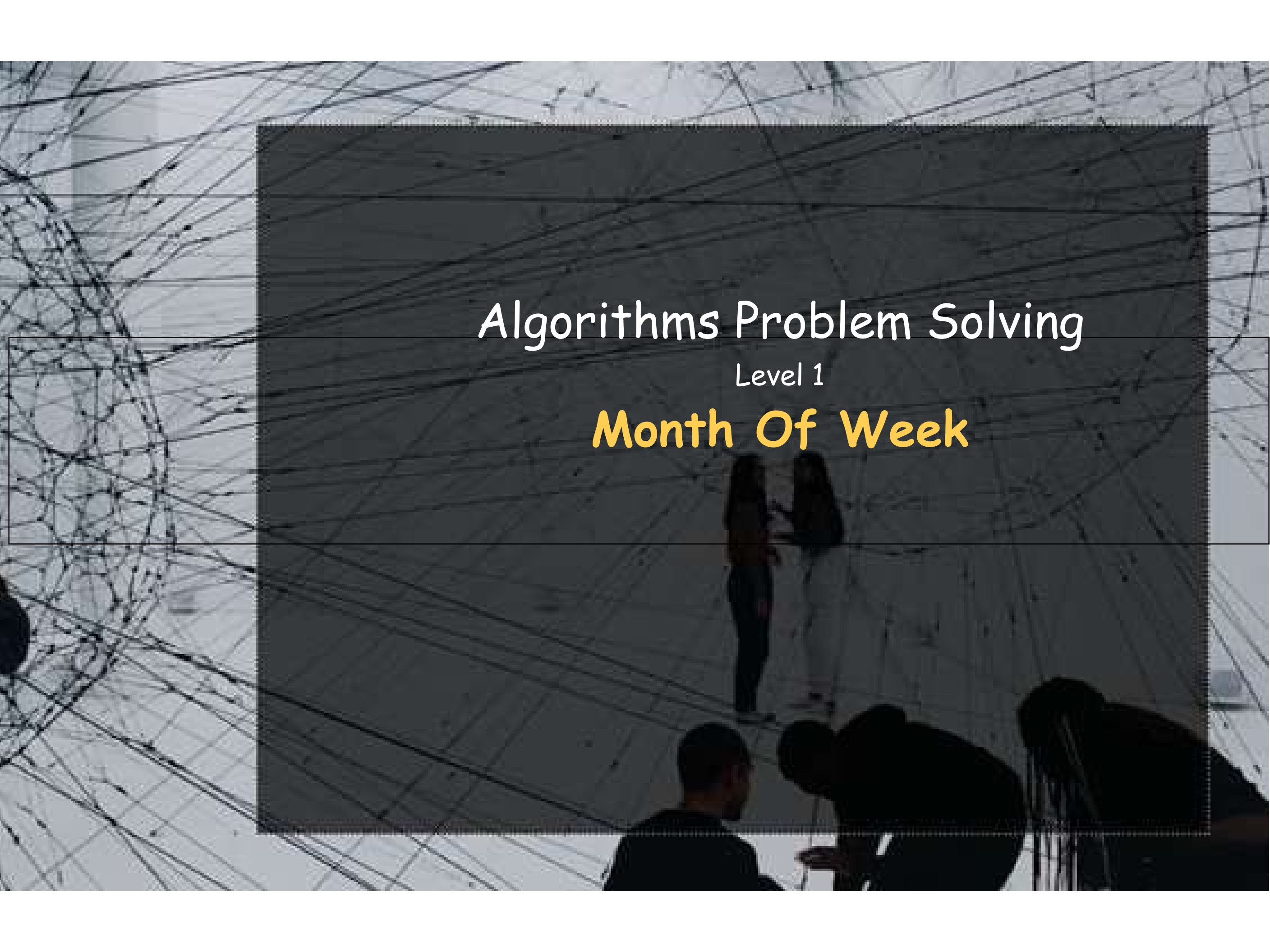


Algorithms Problem Solving

Level 1

Month Of Week



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Definition

- ◆ This program asks the user to enter a month number from 1 to 12 and prints the corresponding month name.
- ◆ The program uses an enum to represent the months and a `switch` statement to select the correct month.
- ◆ If the user enters a number outside 1-12, the program prints an invalid month message.

Explanation

◆ `enMonth` is an enum representing the twelve months:

1 = January

:

:

:

:

12 = December

◆ Since `January = 1` is explicitly assigned, the following months automatically receive the next values.

◆ `ShowMonthsMenu()` displays all months and asks the user to enter a month number.

◆ `ReadMonth()` reads the user's number and converts it to the `enMonth` enum type.

◆ `GetMonthName()` uses a `switch` statement to match the selected month with its name.

◆ The `default` case handles numbers that are not valid month numbers.

◆ In `main()`, the returned month from `ReadMonth()` is passed directly to `GetMonthName()`.

Steps in main()

1. Define the `enMonth` enum.
2. Assign numbers 1-12 to the months.
3. Display the months menu.
4. Read the user's month number.
5. Convert the number to `enMonth`.
6. Use `switch` to identify the month.
7. Return the month name.
8. Print the selected month.

💡 Example

◆ If the user enters:

8

◆ The program matches `8` with:
`August`

➡ Result

Selected month is: August

Note

- ◆ An enum gives meaningful names to numeric values.
- ◆ Because `January = 1`, the months automatically continue from 2 to 12.
- ◆ The `default` case handles invalid values such as 0 or 13.

⊘ Warning

◆ `ReadMonth()` converts any entered integer directly to the enum, so it does not validate the input first. Invalid values are handled later by the `default` case in `GetMonthName()`.



Comparison

▨ Valid Month

- ✓ The entered number is between 1 and 12.
- ✓ It matches one of the enum values.
- ✓ The corresponding month name is returned.

▤ Invalid Month

- ✓ The entered number is outside 1-12.
- ✓ No `case` matches the value.
- ✓ The `default` case returns `Invalid month number!`.

✦✦ Summary

- ✓ An enum can represent the months using meaningful names.
- ✓ `switch` can map each month number to its corresponding name.
- ✓ The `default` case handles invalid month numbers.
- ✓ Enum values can automatically increase when the first value is explicitly assigned.

Thank you 😊